C 40975 (Pages : 4) Name.....

Reg. No.....

FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2023

Mathematics

MAT 4C 04—MATHEMATICS

(2017 and 2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type)

Answer all the **twelve** *questions.*
Each question carries 1 *mark.*

1. Find Wronskian of e^x and xe^x ?
2. Find L (sinh at).
3. Define convolution of two functions f and g
4. Solve $y'' + 6y' + 9y = 0$.
5. State existence and uniqueness theorem for initial value problem.
6. What do you mean by general solution of an ODE ?
7. State Fundamental theorem on superposition.
8. Define basis of solution of an ODE.
9. State existence theorem for Laplace transform.
10. Write the 2-dimensional Poisson equation.
11. What do you mean by an odd function ? Give example.
12. State trapezoidal rule for numerical integration.

(12 × 1 = 12 marks)

Turn over

Part B (Short Answer Type)

Answer any **nine** *questions.*

Each question carries 2 *marks.*

13. Find L $(e^{-kt}(a \cos t + b \sin t))$.

14. Solve $xy'' + 2y' = 0$.

15. Verify that $u = 4x^2 + t^2$ is a solution of the wave equation with suitable c.

16. Show that Inverse Laplace transform is a linear operator.

17. Check whether the functions $e^{-x} \cos wx$, $e^{-x} \sin wx$ are linearly independent.

18. Find the particular integral for $y' + 4y = 8x^2$.

19. Find the Fourier series of $f(x) = \begin{cases} -1, & \text{if } -\pi < x < 0; \\ 1, & \text{if } 0 < x < \pi. \end{cases}$ $f(x + 2\pi) = f(x)$.

20. If $f(x)$ is a periodic function of x of period p, Show that $f(x/b), b \neq 0,$ is a periodic function of x of period pb.

21. If L $(f(t))$ = F (s), and c is any positive constant, show that L $(f(ct))$ = F $(s/c)/c$.

22. Show that a constant function $f(x)$ is periodic with any period but has no fundamental period.

23. Factor and solve $(D + 2I)^2 y = 0$.

24. Use the Trapezoidal Rule with $n = 4$ to estimate $\int_1^2 x^2 \, dx$. Compare the estimate with the exact value.

(9 × 2 = 18 marks)

Part C (Short Essay)

Answer any **six** *questions.*

Each question carries 5 *marks.*

25. Write the following function using unit step functions and find its Laplace transform.

$$f(t)=\begin{cases}2 & \text{if } 0<t<1;\\ \frac{1}{2}t^2, & \text{if } 0<t<\frac{\pi}{2};\\ \cos t, & \text{if } t>\frac{\pi}{2}.\end{cases}$$

26. Solve $y'+\frac{1}{2}y=17\sin 2t, y(0)=-1$ using Laplace transform.

27. Find $\mathrm{L}\left(\sin h^2 at\right)$.

28. Solve $y''-25y=0, y(-2)=y(2)=\cos h10$.

29. Find the Fourier series of $f(t)=\begin{cases}0 & \text{if } -2<x<-1;\\ k, & \text{if } -1<x<1;\\ 0, & \text{if } 1<x<2.\end{cases}$

30. Use the improved Eulers method to calculate the first three approximations to the given initial value problem. Compare the approximations with the values of the exact solution.

$y'=2y(x+1), y(0)=3, dx=0.2.$

31. $y''+5y'+4y=2e^{-2t}, y(0)=0, y'(0)=0$ by using Convolution theorem.

32. Solve $u_{xx}=4y^2u$.

Turn over

33. Estimate the integral $\int_2^4 \frac{1}{(s-1)^2}$ with $n = 4$ steps and find an upper bound for $|E_s|$ using Simpson's rule.

(6 × 5 = 30 marks)

Part D

Answer any **two** *questions.*

Each question carries 10 *marks.*

34. Solve $y'' + 9y = 8 \sin t$ if $0 < t < \pi$ and 0 if $t > \pi$; $y(0) = 0$, $y'(0) = 4$.

35. Show that the identities $\sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3x$ can be interpreted as Fourier series expansions. Develop $\cos^4 x$.

36. Solve the integral equation $y(t) - \int_0^t y(\tau) \sin(t - \tau)\, d\tau = \cos t$.

(2 × 10 = 20 marks)